

Mathematics for Data Science & Machine Learning (Vol. 2)

Comprehensive Practical Question Bank: Questions 201–400

This document serves as Volume 2 of your comprehensive problem-solving math workbook, containing an entirely fresh set of 200 applied math questions across Linear Algebra, Probability, Statistics, and Differential Calculus. Like the previous compilation, this set features realistic numbers, matrices, and datasets arranged progressively through foundational, intermediate, and advanced levels to ground your computational skills for AI applications.

1. Linear Algebra (Questions 201–250)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 201–215)

- 201.** Given vectors $\mathbf{a} = [2, 5, -1]^T$ and $\mathbf{b} = [-3, 0, 4]^T$, compute the linear combination $4\mathbf{a} + 2\mathbf{b}$ and determine its Euclidean (L_2) norm.
- 202.** Calculate the dot product of two embedding layers $\mathbf{u} = [-2, 1, 3, 5]$ and $\mathbf{v} = [3, -4, 2, 0]$. State whether they form an acute, obtuse, or right angle.
- 203.** Find the Manhattan (L_1) and Chebyshev (L_∞) norms of the parameter update discrepancy vector $\Delta\mathbf{w} = [0.15, -0.60, 0.05, 1.10, -0.22]$.
- 204.** Compute the cosine similarity score between two item-recommendation user profile vectors: $\mathbf{x} = [4, 1, 3]$ and $\mathbf{y} = [2, 5, 0]$.
- 205.** Find the resulting matrix-vector product $\mathbf{y} = \mathbf{W}\mathbf{x}$ given:

$$\mathbf{W} = \begin{bmatrix} 1 & 4 & -2 \\ 3 & 0 & 5 \\ -1 & 2 & 3 \end{bmatrix}$$
$$\mathbf{x} = [3, -1, 2]^T$$

- 206.** For the following two dense transformation matrices, compute the matrix product $\mathbf{C} = \mathbf{AB}$:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -1 & 3 \end{bmatrix}$$
$$\mathbf{B} = \begin{bmatrix} 1 & 5 & 0 & 2 \\ -3 & 1 \end{bmatrix}$$

- 207.** Verify if matrix multiplication is commutative for the following pair by evaluating $\mathbf{AB}$ and $\mathbf{BA}$:

$$\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 1 & 5 \end{bmatrix}$$
$$\mathbf{B} = \begin{bmatrix} 0 & 3 \\ 2 & -1 \end{bmatrix}$$

208. Demonstrate that the transpose property $(CD)^T = D^T C^T$ holds exactly using:

$$C = \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix}$$
$$D = \begin{bmatrix} 4 & 0 \\ -2 & 3 \end{bmatrix}$$

209. Find the determinant of the 2x2 projection matrix $M = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$. Is M singular or non-singular?

210. Compute the analytical inverse W^{-1} for the 2x2 layer weight matrix:

$$W = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

211. Find the trace of the square feature covariance matrix Σ given by:

$$\Sigma = \begin{bmatrix} 8.4 & 1.2 & 0.0 \\ 1.2 & 5.1 & -2.3 \\ 0.0 & -2.3 & 9.7 \end{bmatrix}$$

212. Determine if the following three feature vectors are linearly independent or dependent: $x_1 = [1, 0, 2]$, $x_2 = [3, 1, -1]$, and $x_3 = [5, 1, 3]$.

213. Determine the rank of the matrix A by finding its row echelon form:

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 2 & 1 \\ 3 & 6 & 5 \end{bmatrix}$$

214. Project the support vector data point $x = [3, 7]^T$ onto the linear subspace spanned by the unit vector $v = [0.8, -0.6]^T$. Find the scalar projection value.

215. Determine whether the rotation matrix R is an orthogonal matrix by checking if $R^T R = I$:

$$R = \begin{bmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{bmatrix}$$

B. INTERMEDIATE LEVEL (QUESTIONS 216–235)

216. Solve the system $Ax = b$ via Gaussian Elimination representing a three-factor production pipeline:

$$\begin{aligned} x + 2y + 3z &= 9 \\ 2x - y + z &= 8 \\ 3x + y - z &= 3 \end{aligned}$$

217. Find the determinant of the 3x3 scaling matrix using cofactor expansion down the second column:

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 4 & -1 \\ 5 & 2 & 1 \end{bmatrix}$$

218. Find the inverse of the 3x3 matrix $\mathbf{B}$ using the Gauss-Jordan elimination approach:

$$\mathbf{B} = \begin{bmatrix} 1, & 2, & 0 \\ 0, & 1, & 3 \\ 2, & 0, & 1 \end{bmatrix}$$

219. Compute the exact eigenvalues and corresponding eigenvectors for the 2x2 matrix $\mathbf{M} = \begin{bmatrix} 6, & 2 \\ 2, & 3 \end{bmatrix}$.

220. Given the symmetric matrix $\mathbf{A} = \begin{bmatrix} 4, & 2 \\ 2, & 1 \end{bmatrix}$, determine its orthogonal diagonalization components $\mathbf{P}$ and $\mathbf{D}$.

221. Compute the Frobenius norm of the given 3x2 parameter weight discrepancy matrix $\Delta \mathbf{W}$:

$$\Delta \mathbf{W} = \begin{bmatrix} 0.2, & 1.5 \\ -0.4, & -0.8 \\ 1.0, & 0.5 \end{bmatrix}$$

222. Given a small matrix $\mathbf{X}$ of 3 observations and 2 features, compute the uncentered Gram matrix $\mathbf{G} = \mathbf{X}\mathbf{X}^T$:

$$\mathbf{X} = \begin{bmatrix} 1, & 3 \\ 2, & 1 \\ 0, & 4 \end{bmatrix}$$

223. Test if the quadratic form $q(\mathbf{x}) = 2x_1^2 - 4x_1x_2 + 5x_2^2$ is positive definite using Sylvester's criterion on principal minors.

224. A vector has coordinates $\mathbf{x}_B = [3, 2]^T$ relative to the basis $\mathbf{B} = \{[2, 1]^T, [-1, 3]^T\}$. Determine its coordinates in the standard basis.

225. Compute the Hadamard element-wise product $\mathbf{U} \odot \mathbf{V}$ and the standard matrix product $\mathbf{UV}$ for the dropout masks:

$$\begin{aligned} \mathbf{U} &= \begin{bmatrix} 1, & 0 \\ 1, & 1 \end{bmatrix} \\ \mathbf{V} &= \begin{bmatrix} 0.5, & 2 \\ 0, & 1.5 \end{bmatrix} \end{aligned}$$

226. Apply the Gram-Schmidt process to transform the linearly independent vectors $\mathbf{u}_1 = [1, 0, 1]$ and $\mathbf{u}_2 = [-1, 2, 0]$ into an orthonormal basis.

227. Determine the basis and dimension of the null space (kernel) for the system matrix:

$$\mathbf{A} = \begin{bmatrix} 1, & -2, & 3 \\ 2, & -4, & 6 \\ -1, & 2, & -3 \end{bmatrix}$$

228. Compute the condition number $\kappa(\mathbf{A})$ in the L_∞ norm for the system matrix:

$$\mathbf{A} = \begin{bmatrix} 2, & 1 \\ 2, & 1.005 \end{bmatrix}$$

229. Find the LU Decomposition ($A = LU$) without pivoting for the matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$$

230. A directional network has the adjacency matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Compute A^2 to map the number of 2-step connections between nodes.

231. Find the basis for the row space and column space of the following matrix:

$$M = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 0 & 2 \\ 4 & 2 & 6 \end{bmatrix}$$

232. Compute the Moore-Penrose pseudo-inverse A^+ for the single-row matrix $A = [3, 4]$.

233. Express vector $v = [1, 5]$ as a linear combination of the alternative coordinate vectors $e_1 = [3, 1]$ and $e_2 = [1, 2]$.

234. An affine operator scales a vector by 2 along the x-axis and translates it by vector $b = [-1, 4]^T$. Write this as $Ax + b$ and evaluate it for $x = [3, 2]^T$.

235. Show that if a matrix P satisfies $P = P^T$ and $P = P^2$, then the vector $(I - P)x$ is orthogonal to Px for any vector x .

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 236–250)

236. Compute the Singular Value Decomposition (SVD), $A = U\Sigma V^T$, for the rectangular matrix:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$$

237. A zero-mean centered dataset matrix X contains 4 samples across 2 features:

$$X = \begin{bmatrix} -2 & 1 \\ -1 & -1 \\ 1 & 0 \\ 2 & 0 \end{bmatrix}$$

Compute the empirical covariance matrix $\Sigma = (1/4)X^T X$ and determine its principal component loading directions.

238. Find the optimal least-squares parameter solution vector $w = (X^T X)^{-1} X^T y$ for the design matrix:

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \\ y = [1, 3, 4]^T$$

239. Project the data point $x = [4, 1]$ onto the top principal component direction of the covariance matrix $\Sigma = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$.

240. Compute the Rank-1 optimal low-rank matrix approximation A_1 via truncated SVD for:

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$$

241. Find the Mahalanobis distance between sample point $x = [3, 2]^T$ and cluster center $\mu = [1, 2]^T$ given the inverse covariance structure:

$$\Sigma^{-1} = \begin{bmatrix} 1.2 & -0.4 \\ -0.4 & 0.8 \end{bmatrix}$$

242. Given the graph adjacency matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and its corresponding diagonal degree matrix D , compute the unnormalized graph Laplacian matrix $L = D - A$.

243. Derive the gradient vector with respect to w for the weighted ridge constraint function $f(w) = w^T D w$, where D is a constant diagonal matrix.

244. Compute the Moore-Penrose pseudo-inverse matrix X^+ for the full-column-rank design matrix:

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

245. Prove that the eigenvalues of a symmetric positive definite matrix must all be strictly positive real numbers.

246. A steady-state page routing probability satisfies $vP = v$. Find the normalized vector v given the stochastic matrix:

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

247. Using a Radial Basis Function (RBF) kernel matrix calculation $K(x, y) = \exp(-\gamma \|x - y\|^2)$ with $\gamma = 0.5$, calculate the kernel similarity score between points $x = [1, 3]$ and $y = [2, 1]$.

248. Determine the structural whitening transformation matrix $W = \Sigma^{-1/2}$ for a dataset whose current covariance is $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$.

249. Compute the Kronecker product $A \otimes B$ for the mini-matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \\ B = \begin{bmatrix} 2 & 1 \\ 4 & 0 \end{bmatrix}$$

250. In an optimization context, show that the spectral radius (maximum absolute eigenvalue) of the iteration matrix B must be strictly less than 1 for the linear numerical system $x_{k+1} = Bx_k + c$ to converge to its fixed-point solution.

2. Probability (Questions 251–300)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 251–265)

251. An API gateway routes 70% of transactions to Service Cluster X and 30% to Cluster Y. Service Cluster X errors out on 1% of transactions, while Cluster Y errors out on 4%. Find the total probability that an incoming transaction errors out.

252. Given probabilities $P(A) = 0.6$, $P(B) = 0.3$, and $P(A \cap B) = 0.18$, compute $P(A | B)$ and state whether events A and B are independent.

253. In a financial fraud application, 2% of accounts are fraudulent. A rule-based alert flags 90% of fraudulent accounts and 5% of legitimate accounts. Calculate the posterior probability that an account flagged by the alert is actually fraudulent.

254. A discrete random feature score X displays the following PMF:

$X:$	$[0,$	$1,$	$5,$	$10]$
$P(X):$	$[0.2,$	$0.5,$	$0.2,$	$0.1]$

Find the expected value $E[X]$.

255. For the random variable X detailed in Question 254, compute the variance $\text{Var}(X)$ and standard deviation σ .

256. A model conversion rate is $p = 0.3$. Using the Binomial distribution formula, calculate the probability of securing exactly 3 conversions out of 8 independent user clicks.

257. The number of exceptions thrown by an application tracking thread follows a Poisson distribution averaging 1.5 per hour. Compute the probability of seeing exactly 3 exceptions in a given hour.

258. A data cleaning delay X is uniformly distributed between 5 and 25 seconds. Find the probability that a pipeline task requires between 10 and 22 seconds.

259. Engine cluster temperatures are normally distributed with $\mu = 120$ degrees and $\sigma = 15$ degrees. Calculate the standard Z-score corresponding to a sensor reading of 145 degrees.

260. Using standard normal lookup statistics where $\Phi(1.67) = 0.9525$, find the probability that a randomly sampled engine cluster runs at a temperature below 145 degrees based on the parameters in Question 259.

261. A data scuffing script has a probability $p = 0.05$ of extracting a damaged record. What is the expected number of records parsed before hitting the first damaged row?

262. If three independent testing stages have individual success probabilities of 0.8, 0.9, and 0.75, what is the probability that all three stages complete successfully?

263. Express the statistical Recall metric of a classification test as a conditional probability involving true positive states and label space states.

264. A network router drops packets with a 0.5% probability. Find the probability that the first dropped packet occurs on exactly the 10th packet transmission.

265. Prove the Law of Total Probability for a sample space partitioned into three mutually exclusive scenarios B_1 , B_2 , B_3 .

B. INTERMEDIATE LEVEL (QUESTIONS 266–285)

266. A sensor-based health diagnostic test displays 95% sensitivity and 98% specificity. The baseline condition rate in the target cohort is 1%. If an individual tests positive, evaluate the probability they have the underlying condition.

267. Two random variable discrete metrics X and Y reveal the following joint PMF grid:

	$Y=1$	$Y=2$
$X=1$	0.15	0.25
$X=2$	0.35	0.25

Compute the individual marginal PMF values $P(X)$ and $P(Y)$.

268. Using the joint table in Question 267, compute the conditional expectation value $E[X | Y=1]$.

269. Calculate the exact covariance $Cov(X, Y)$ using the joint distribution provided in Question 267.

270. Determine the Pearson correlation coefficient $\rho_{X,Y}$ for the features described in Question 267.

271. The time required to process a queue order follows an exponential layout with a mean duration of 4 minutes ($\lambda = 0.25$). Calculate the probability that an order takes more than 6 minutes.

272. Given two uncorrelated random feature adjustments A and B with $Var(A) = 16$ and $Var(B) = 25$, determine the variance of the linear contrast score $Z = 2A - 3B$.

273. A continuous density shape operates via $f(x) = k(1 - x)$ for $0 \leq x \leq 1$, and 0 elsewhere. Determine the proper normalization constant k .

274. For the density curve established in Question 273, find the expected value $E[X]$.

275. Derive the explicit Cumulative Distribution Function expression $F(x) = P(X \leq x)$ for the density in Question 273.

276. User connection dropouts follow a Poisson pattern at an average rate of 0.2 dropouts per minute. Find the probability of noting zero dropouts during a 10-minute web monitoring test.

277. A distributed node relies on three independent backup hard loops, each with a survival probability of 0.85. The system remains operational if at least one loop survives. Find the overall reliability.

278. Using the Central Limit Theorem, approximate the probability that the average score of 40 independent uniform samples drawn from $U(0, 10)$ is greater than 5.5. (For $U(0,10)$, $\mu = 5$, $\sigma^2 = 100/12$).

279. Prove analytically that the covariance between two variables satisfies $Cov(X, Y) = E[XY] - E[X]E[Y]$.

280. Let independent latent layers be modeled as $A \sim N(5, 2)$ and $B \sim N(12, 7)$. Define the parameters for the distribution of their difference $W = B - A$.

281. A classifier model has a 10% error rate on test sets. If you sample 6 test items at random, what is the probability that at most 1 item is misclassified?

282. Assume conditional profiles indicate $P(X \mid \text{Class A}) = 0.7$ and $P(X \mid \text{Class B}) = 0.4$. If the prior probability of Class A is 0.3 and Class B is 0.7, calculate the posterior probability $P(\text{Class A} \mid X)$.

283. An extraction box holds 8 clean items and 4 corrupted items. If you draw 3 items without replacement, find the probability that exactly 1 corrupted item is chosen.

284. Compute the Shannon entropy in bits for a 4-state multinomial categorical feature with probabilities $p = [0.5, 0.25, 0.125, 0.125]$.

285. Apply Chebyshev's Inequality to find the maximum possible probability that a random feature observation deviates from its mean by more than 3 standard deviations.

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 286–300)

286. A bivariate normal feature pair $[X, Y]^T$ has mean vector $\mu = [1, 2]^T$ and covariance matrix $\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Write the complete analytical algebraic density function expression $f(x, y)$.

287. For the bivariate distribution in Question 286, derive the explicit conditional distribution parameters for Y given that $X = 2$.

288. Compute the Kullback-Leibler (KL) Divergence $D_{KL}(P \parallel Q)$ between target distribution $P = [0.8, 0.2]$ and model prediction $Q = [0.7, 0.3]$.

289. Prove that the KL divergence between two multivariate normal distributions $N_0(\mu_0, \Sigma_0)$ and $N_1(\mu_1, \Sigma_1)$ reduces to a trace and quadratic form expression when $\Sigma_0 = \Sigma_1 = I$.

290. In a document multi-label framework, write down the joint probability formulation for a text item containing three conditionally independent features given the parent theme class label.

291. Derive the Moment Generating Function (MGF), $M_X(t) = E[e^{tX}]$, for a continuous random variable following an Exponential distribution with parameter λ .

292. Using the Inverse CDF transformation rule, derive a sampling expression to generate random draws from a Pareto distribution with minimum scale x_m and shape parameter α , given standard uniform variables $U \sim \text{Uniform}(0,1)$.

293. A Gaussian mixture profile tracks three clusters with weights $\pi = [0.2, 0.5, 0.3]$. Given their individual component densities are $p_1(x)$, $p_2(x)$, and $p_3(x)$, write the complete marginal density equation for an observation x .

294. Derive the variance matrix expression for a linear mapping vector transformation $Y = AX$, where X is a random vector with covariance matrix Σ and A is a constant multiplier matrix.

295. In Bayesian model estimation, if the data likelihood follows a Poisson distribution, identify the conjugate prior distribution for the rate parameter λ .

296. A Markov clickstream process runs on three states. The transition probability matrix is:

$$P = \begin{bmatrix} 0.5 & 0.5 & 0.0 \\ 0.1 & 0.6 & 0.3 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

Given an initial user probability state vector $\mathbf{v}_0 = [0.0, 1.0, 0.0]$, compute the probability vector after 2 step transitions ($\mathbf{v}_2$).

297. Apply the union bound and Hoeffding's Inequality to express a general generalization error bound over a finite hypothesis space $|\mathbf{H}|$ containing M models.

298. Write out the joint probability distribution definition for a Dirichlet distribution with parameter vector $\alpha = [a_1, a_2, a_3]^T$.

299. State the explicit optimization objective function for the Evidence Lower Bound (ELBO) used to approximate posterior fields in Variational Autoencoders (VAEs).

300. Let $Z_1, Z_2, \dots, Z_k$ be k independent standard normal variables. If you define a new metric $W = \sum Z_i^2$, name the resulting distribution family and state its analytical mean and variance.

3. Statistics (Questions 301–350)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 301–315)

301. Calculate the sample mean and median for the following set of model cross-validation training times (in seconds):

Data: [12.4, 15.1, 11.8, 24.5, 14.2, 13.0, 16.5]

302. Compute the sample variance and sample standard deviation for the training duration observations listed in Question 301. Use Bessel's Correction ($n-1$).

303. Find the range and the Interquartile Range (IQR) for the following listing of deep network feature counts per layer:

Data: [64, 64, 72, 80, 96, 128, 128, 256, 1024]

304. Identify if the final value (1024) in the feature counts array of Question 303 constitutes an outlier using the standard $1.5 \times IQR$ rule.

305. A dataset sample of size $n = 100$ gives a sample mean $\bar{x} = 42.5$. Given a known population standard deviation $\sigma = 5.0$, establish the 95% confidence interval for the true population mean.

306. Formulate the explicit Null and Alternative hypotheses required to test if reducing data resolution compromises model prediction accuracy.

307. Define Type I and Type II errors in the context of a automated spam-detection filter.

308. An A/B test comparing user engagement returns a p-value of 0.008. Interpret this result relative to a significance threshold of $\alpha = 0.01$.

309. Compute the standard error of the mean (SEM) for a metric profile where the sample standard deviation is $s = 2.45$ and the sample size is $n = 64$.

310. A categorical survey records the following customer sentiment responses across three classes. Compute the expected counts under the null hypothesis that responses are uniformly distributed:

Satisfied: 65, Neutral: 40, Dissatisfied: 45

311. Determine the directional skewness for a dataset where the calculated mean is 85.2, the median is 74.1, and the mode is 62.0.

312. Write out the mathematical steps required to assemble a sample covariance matrix for a two-dimensional feature array:

Feature 1: [1, 3, 5]
Feature 2: [2, 6, 7]

313. Explain how the Central Limit Theorem allows you to perform hypothesis tests on the mean of an exponentially distributed population.

314. A 95% confidence interval for a model's F1-score is calculated as $[0.81, 0.87]$. State the precise frequentist definition of this confidence interval.

315. What is the degrees of freedom for a sample variance estimator, and why do we divide by $n-1$ instead of n ?

B. INTERMEDIATE LEVEL (QUESTIONS 316–335)

316. Perform a one-sample t-test: A baseline classification model takes 120ms to evaluate an image batch. A modified model version runs $n = 16$ trials, yielding a mean latency of **115ms** with $s = 8ms$. Calculate the t-statistic and state your conclusion for a two-tailed test at $\alpha = 0.05$ ($t_{crit} = 2.131$).

317. An independent two-sample t-test with equal variances compares two training initializations. Given the following data, compute the pooled variance s_p^2 and the resulting t-statistic:

Init A: $n_1 = 10$, $\bar{x}_1 = 0.78$, $s_1 = 0.03$
Init B: $n_2 = 12$, $\bar{x}_2 = 0.74$, $s_2 = 0.04$

318. Construct a 99% confidence interval for the true difference between the initialization means using the data from Question 317 (assume $t_{crit} \approx 2.845$ for $df = 20$).

319. Calculate the Chi-Square (χ^2) test statistic for the following 2x2 contingency table tracking user conversions across two model recommendations:

Recommender	Converted	Ignored
Model X	80	120
Model Y	110	90

320. Using the Chi-Square statistic from Question 319, determine if the choice of recommender significantly affects conversions at $\alpha = 0.05$ (Critical χ^2 for $df=1$ is **3.841**).

321. Compute the exact Pearson correlation coefficient r between feature X and target label Y from the following small training subset:

$X = [2, 4, 6, 8]$
 $Y = [3, 5, 8, 10]$

322. Fit an ordinary least squares (OLS) linear trend line $\hat{y} = \beta_0 + \beta_1 x$ using the small training sample from Question 321.

323. For the linear model derived in Question 322, compute the coefficient of determination R^2 by calculating the ratio of the explained sum of squares (ESS) to the total sum of squares (TSS).

324. Explain the relationship between the statistical power of an experiment and the probability of making a Type II error. How does increasing sample size affect power?

325. A cluster response latency shows a log-normal profile with a mean log-latency of 3.5 and a log standard deviation of 0.4. Find the 95th percentile response latency threshold.

326. Convert the following coordinate pairs into a Spearman's rank correlation coefficient to assess non-linear monotonic correspondence:

$X = [10, 25, 18, 40]$
 $Y = [100, 450, 200, 800]$

327. When analyzing user ratings that are highly skewed and ordinal, why is the Wilcoxon signed-rank test preferred over a standard paired t-test?

328. Complete the missing values in this one-way ANOVA framework. You are evaluating four model frameworks across 5 validations each. Given Between-Group Sum of Squares (SSB) = 60 and Within-Group Sum of Squares (SSW) = 80, compute the Mean Square fields (MSB, MSW) and the final F-statistic.

329. Calculate Cohen's d for the experiment described in Question 316 to determine the effect size of the modified model version.

330. A sample size of $n = 25$ yields a sample mean of 15.0 and a sample standard deviation of 2.0. Construct the 95% confidence interval for the population mean using the t-distribution ($t_{0.025, df=24} = 2.064$).

331. Describe how the shape of the Student's t-distribution changes relative to the standard normal distribution as the degrees of freedom approach infinity.

332. You run 50 parallel independent A/B tests on a website using a standard significance level of $\alpha = 0.05$. If the null hypothesis is true for all tests, what is the expected number of false discoveries? Apply the Bonferroni adjustment to find the corrected alpha threshold.

333. Determine the numerator and denominator degrees of freedom for an F-test within an ANOVA design that compares 3 algorithms evaluated across 15 distinct datasets each.

334. Define consistency as a property of an estimator. Is the sample variance s^2 a consistent estimator for the population variance σ^2 ?

335. Write out the probability density function for the F-distribution and describe its structural role in testing variance equality between two populations.

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 336–350)

336. Let $x_1, x_2, \dots, x_n$ be i.i.d. random observations sampled from a Geometric distribution with success parameter p . Derive the Maximum Likelihood Estimator (MLE) for p .

337. A user interacts with an ad portal, clicking the ad on the 4th, 6th, 3rd, and 5th attempts across four independent sessions. Calculate the numerical MLE estimate for the click probability p based on your derivation in Question 336.

338. Derive the Maximum Likelihood Estimator for the variance parameter σ^2 of a normal distribution $N(\mu, \sigma^2)$ given n i.i.d. observations, assuming the mean μ is known.

339. Derive the Maximum A Posteriori (MAP) estimator for the mean parameter μ of a normal distribution given a normal prior $\mu \sim N(\mu_0, \sigma_0^2)$ and n i.i.d. data points. Show how the MAP estimate balances prior knowledge with the empirical data.

340. In a regression context, prove that the Mean Squared Error (MSE) can be decomposed into the sum of the square of the model bias, the model variance, and the irreducible noise variance.

341. At a specific feature point x , a regression model has a true target value $y = 10.0$. Across several random training iterations, the model's predictions have an expected value of $E[\hat{y}] = 9.2$ and a variance of $\text{Var}(\hat{y}) = 0.4$. If the irreducible error variance is $\sigma^2 = 0.25$, calculate the total expected MSE at x .

342. Outline the step-by-step procedure for using the Bootstrap method to construct a 95% percentile confidence interval for the regularized weights of a Ridge regression model.

343. A logistic regression model has parameter weights $w = [1.5, -0.5]^T$ and bias $b = -1.0$. For an input vector $x = [2.0, 4.0]^T$, calculate the log-odds and the corresponding predicted probability for the positive class.

344. Derive the Fisher Information matrix for a normal distribution's mean parameter μ given n i.i.d. observations, assuming the variance σ^2 is constant.

345. Explain how changing the regularizer coefficient λ in Ridge regression scales the bias and variance of the model's parameters.

346. Describe the primary assumptions of classical ordinary least squares (OLS) linear regression. What diagnostic plots or statistics can you use to check the independence assumption?

347. Calculate the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for a regularized neural network containing $k = 12$ parameters evaluated over a sample of $n = 500$ observations, given a maximized log-likelihood value of $\ln L = -150.2$.

348. Why does evaluating a model using K-fold cross-validation provide a better estimate of generalization performance than evaluating it on the training set? Explain how cross-validation helps mitigate overfitting.

349. If a regression model exhibits severe heteroscedasticity, how does this affect the validity of standard OLS hypothesis tests on the coefficients? Suggest a transformation technique to address this issue.

350. Contrast the Benjamini-Hochberg procedure with the Bonferroni correction. Why is the Benjamini-Hochberg procedure generally preferred when performing high-dimensional feature selection across thousands of candidate variables?

4. Differential Calculus (Questions 351–400)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 351–365)

351. Compute the first derivative of the polynomial function: $f(x) = -2x^4 + 6x^3 - 7x + 10$.

352. Find the first derivative of the log-scaled function: $f(x) = \ln(x^2 + 2)$ for all real numbers x .

353. Apply the Product Rule to find the derivative of the combined function: $f(x) = x^3 \ln(x)$ for $x > 0$.

354. Apply the Quotient Rule to find the derivative of the function: $f(x) = e^x / (x^2 + 1)$.

355. Using the Chain Rule, find the explicit derivative of the composite activation function: $f(x) = \exp(-3x^2)$.

356. Find the slope of the tangent line to the curve $f(x) = x^3 - 2x^2 + 4$ at the coordinate point $x = 1$.

357. Evaluate the first derivative of the modified Sigmoid activation function $g(x) = 1 / (1 + e^{-2x})$. Express your answer in terms of $g(x)$.

358. Find the partial derivatives $\partial f / \partial x$ and $\partial f / \partial y$ for the multi-variable function: $f(x, y) = 2x^3y^2 - 4x^2y^4$.

359. Compute the gradient vector $\nabla f(x, y)$ for the function $f(x, y) = \ln(x^2 + y^2)$ at the operational evaluation point $(1, 2)$.

360. Identify all critical points of the function $f(x) = 2x^3 - 9x^2 + 12x - 3$ and classify each as a local minimum or maximum.

361. Compute the second derivative $f''(x)$ of the function $f(x) = x e^{-x}$. Find the point where the curve changes concavity ($f''(x) = 0$).

362. Find the derivative of the Exponential Linear Unit (ELU) activation function given by $f(x) = x$ if $x > 0$ and $f(x) = 0.5(e^x - 1)$ if $x \leq 0$. Discuss its continuity at $x = 0$.

363. Apply L'Hôpital's Rule to evaluate the numerical limit of the indeterminate ratio: $\lim_{(x \rightarrow 0)} ((e^x - 1) / x)$.

364. Compute the directional derivative of $f(x, y) = 2x^2y^3$ at the point $(2, 1)$ in the direction of the unit vector $u = [1/\sqrt{2}, 1/\sqrt{2}]^T$.

365. Write out the first-order Taylor series approximation formula for a multi-variable function $f(x, y)$ centered around a localized pivot point (a, b) .

B. INTERMEDIATE LEVEL (QUESTIONS 366–385)

366.

Find the gradient vector ∇f for the Mean Absolute Error (MAE) loss approximation function for a single sample: $f(\mathbf{w}_1, \mathbf{w}_2) = \sqrt{((\mathbf{w}_1 x_1 + \mathbf{w}_2 x_2) - y)^2 + \epsilon}$ with respect to the weight vector $\mathbf{w} = [\mathbf{w}_1, \mathbf{w}_2]$.

367. Evaluate the gradient of the function from Question 366 using the data values: $x_1 = 1, x_2 = 3, y = 5, \epsilon = 0.01$, evaluated at current weights $\mathbf{w}_1 = 2, \mathbf{w}_2 = 1$.

368. Perform one step of Gradient Descent to update the weight vector $\mathbf{w}$. Use the current weights and gradient vector from Question 367 with a learning rate parameter $\eta = 0.5$.

369. Compute the 2x2 Hessian matrix $\mathbf{H}$ of second-order partial derivatives for the multi-variable optimization profile: $f(x, y) = x^3 + 2xy + y^2$.

370. Analyze the eigenvalues of the Hessian matrix calculated in Question 370 evaluated at the critical point $(1/3, -1/3)$ to determine its stability type.

371. Find the derivative of the Swish activation function, $f(x) = x \sigma(x)$, where $\sigma(x)$ is the standard Sigmoid function. Express your solution in terms of $\sigma(x)$ and $f(x)$.

372. Compute the derivative of the categorical cross-entropy loss function for a single sample class index: $L = -y_i \ln(p_i)$ with respect to the model's predicted probability output p_i .

373. Using the chain rule, compute the derivative of a regularized loss function $L_{reg} = L_0 + \alpha|\mathbf{w}|$ with respect to weight $\mathbf{w}$ for $\mathbf{w} \neq 0$, given $\frac{\partial L_0}{\partial \mathbf{w}} = \mathbf{G}$.

374. Find the second-order Taylor series approximation for the exponential activation function $f(x) = e^x$ centered around the evaluation pivot $a = 0$.

375. Use implicit differentiation to find the derivative expression dy/dx along the hyperparameter boundary curve defined by $x^2 y + y^3 = 10$.

376. Compute the Jacobian matrix $\mathbf{J}$ for a coordinate coordinate mapping vector function: $F(x, y) = [\ln(x) + y, e^{xy}]^T$.

377. Evaluate the Jacobian matrix $\mathbf{J}$ from Question 376 numerically at the sample coordinates $(x=1, y=2)$.

378. Find the global minimum of the regularized cost function $L(\mathbf{w}) = (\mathbf{w} - 5)^2 + \lambda|\mathbf{w}|^2$ by optimizing analytically with respect to $\mathbf{w}$ when $\lambda = 0.25$.

379. Given the vector function $f(\mathbf{x}) = \|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x}$, write down the vector derivative expression for $\nabla_{\mathbf{x}} f(\mathbf{x})$.

380. State the definition of strict convexity for a multivariate function. If a function's Hessian matrix is positive semi-definite everywhere but not strictly positive definite, can a critical point still be a unique global minimum?

381.

Compute the derivative of the Parametric ReLU (PReLU) activation function: $f(x) = \max(\alpha x, x)$ for both $x > 0$ and $x < 0$, where α is a learnable parameter.

- 382.** Find the partial derivative of the GELU (Gaussian Error Linear Unit) approximation function $f(x) = 0.5x(1 + \tanh(\sqrt{2/\pi}(x + 0.044715x^3)))$ with respect to x at $x = 0$.
- 383.** Determine the critical points of the function $f(x, y) = x^4 + y^4 - 4xy$ and classify them using the Second Derivative Test.
- 384.** Explain how momentum modifies the standard gradient descent update formula. Write out the coupled system of equations for the velocity vector $\mathbf{v}_{t+1}$ and parameter vector $\mathbf{w}_{t+1}$.
- 385.** Compute the partial derivative of the softmax function $f(x_1, x_2) = e^{-x_1} / (e^{-x_1} + e^{-x_2})$ with respect to x_1 .

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 386–400)

- 386.** Using the method of Lagrange Multipliers, maximize the utility function $f(x, y) = xy$ subject to the circular constraint $x^2 + y^2 = 8$ in the first quadrant. Find the optimal values for x, y , and the multiplier λ .
- 387.** Consider a neural network computation path where an error term δ_l at layer l is propagated backward. Derive the backpropagation matrix formulation for the error vector δ_{l-1} at the previous layer given the weight matrix $\mathbf{W}_l$ and the activation derivative matrix $\text{diag}(\sigma'(z_{l-1}))$.
- 388.** For the multi-class Softmax activation vector where $a_i = e^{z_i} / \sum_k e^{z_k}$, and a cross-entropy loss function $L = -\sum_i y_i \ln(a_i)$, show that the total derivative of the loss with respect to the logit z_j simplifies beautifully to the error form: $\partial L / \partial z_j = a_j - y_j$.
- 389.**
Derive the gradient with respect to matrix $\mathbf{W}$ for the linear layer transformation function $f(\mathbf{W}) = \text{Tr}(\mathbf{X}\mathbf{W}\mathbf{X}^T)$ where $\mathbf{X}$ is a constant data matrix.
- 390.** Derive the gradient of the binary cross-entropy loss function summed over a mini-batch of N samples: $L(\mathbf{w}) = -(1/N) \sum [y_n \ln(\sigma(\mathbf{w}^T \mathbf{x}_n)) + (1 - y_n) \ln(1 - \sigma(\mathbf{w}^T \mathbf{x}_n))]$ with respect to the model weight vector $\mathbf{w}$.
- 391.** Using the gradient vector from Question 390, derive the complete batch Hessian matrix expression with respect to $\mathbf{w}$. Show how this matrix can be written in the elegant form $\mathbf{X}^T \mathbf{R} \mathbf{X}$ where $\mathbf{R}$ is a diagonal weight matrix.
- 392.** Explain how the Levenberg-Marquardt optimization algorithm acts as a bridge between the Gauss-Newton method and standard Gradient Descent. Write down its regularized system update equation.
- 393.** Apply matrix differential rules to find the gradient of the log-likelihood function for a multivariate normal distribution with respect to its precision matrix (the inverse covariance matrix $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$).
- 394.** Formulate the primal and dual optimization problems for a Support Vector Machine (SVM) classifier separating data with a soft-margin parameter C . Explicitly state the complementary slackness conditions.
- 395.** Derive the gradient of the trace of an inverted matrix function, $f(\mathbf{A}) = \text{Tr}(\mathbf{A}^{-1}\mathbf{B})$, with respect to the matrix parameter $\mathbf{A}$.
- 396.** In an LSTM cell, the cell state updates via $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t$. Derive the partial derivative matrix $\partial \mathbf{c}_t / \partial \mathbf{c}_{t-1}$. Use this to explain how the forget gate $\mathbf{f}_t$ prevents vanishing gradients over long temporal sequences.

- 397.** Find the total differential dL for a system loss profile with structural dependency: $L(u, v) = \ln(u^2 + v)$ where $u = x e^y$ and $v = x^2 - y$.
- 398.** Given the scalar cost function $J(W) = 0.5 \|WX - Y\|_F^2$ where W, X, Y are matrices, find the matrix gradient $\nabla_W J(W)$.
- 399.** Prove that for a twice-differentiable multivariate function, a critical point is a strict local minimum if the gradient vector is zero and its Hessian matrix is strictly positive definite.
- 400.** For an optimization setup using AdaGrad, the parameter updates via $w_{t+1} = w_t - (\eta / \sqrt{G_t + \epsilon}) \odot g_t$, where $G_t = \sum_{\tau=1}^t g_\tau^2$. Show how the historical accumulation of partial derivatives modifies the effective local learning rate over long training runs.